

DSA Day 20: Binary Tree Traversals (recursive)

Book: Data Structures Through C in Depth - Ch6 §6.9 · pp 186-188 | Code in Python

Overview

Inorder (L,root,R), preorder (root,L,R), postorder (L,R,root). Inorder of a BST is sorted.

Key concepts to master

- Three visit orders
- Recursive definition

Python implementation

```
def inorder(n):  
    if n: inorder(n.left); print(n.data); inorder(n.right)  
def preorder(n):  
    if n: print(n.data); preorder(n.left); preorder(n.right)
```

Complexity

$O(n)$ time, $O(h)$ stack.

Common pitfalls

- Swapping the order of the three actions
- Forgetting the None check

Practice

- Traverse a tree all 3 ways by hand
- Verify inorder of a BST is sorted

